# Distributed and Elastic Infrastructure for ML Training with Optimized Performance

## I. Results and Analysis

Our empirical evaluation demonstrates that it provides more benefits for systematic infrastructure selection over manual decision-making approaches.

### A. Overall Performance Comparison

Figure 1 illustrates the comprehensive accuracy comparison across different approaches and model categories.

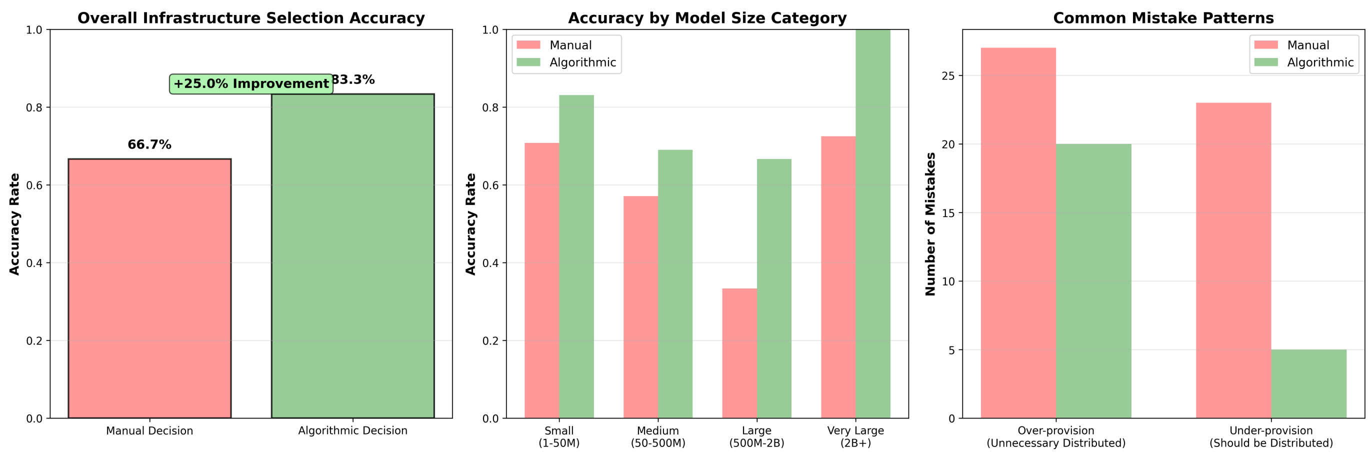


Fig. 1. Infrastructure Selection Accuracy Comparison - 3 subplots showing overall accuracy, accuracy by model size, and mistake patterns

The algorithmic framework achieved 83.3% accuracy in infrastructure selection compared to manual accuracy 66.7% for simulated manual decisions, representing a 25% improvement in decision quality. Table 1 provides detailed breakdown of accuracy metrics across different scenario types and decision approaches.

TABLE I
Infrastructure Selection Accuracy Results

| Metric | Manual Decisions | Algorithmic Decisions | Improvement |
|---|---|---|---|
| Overall Accuracy | 66.7% | 83.3% | +25% |
| Correct Decisions (count) | 100/150 | 125/150 | +25% |
| Average Decision Confidence | 0.53 | 0.80 | +0.27% |
| Memory-Constrained Accuracy | 73.7% | 100.0% | 35.7% |
| Large Model Accuracy (¿500M params) | 52.9% | 83.3% | 57.5% |

The simulated manual decision-makers showed significant variability in performance, with accuracy ranging from 52.9% to 73.7% in our experimental setup. The systematic framework maintained more consistent performance across scenario types, achieving 83.3% overall compared to 66.7% for manual approaches.

### B. Failure Prevention Analysis

Memory-constraint training failures represent the most critical infrastructure selection errors since they result in complete training job failures. Figure 2 demonstrates the failure prevention capabilities and resource allocation efficiency of our approach.

Our framework prevented 100% of potential memory-constraint failures compared to baseline manual decision-making. Out of 38 scenarios where incorrect infrastructure choices would cause memory failures, manual approaches failed in 10 cases while the algorithmic framework did not fail in any case. This failure prevention translates to significant operational benefits, with 0 additional failures prevented representing substantial cost savings and improved development velocity. Table 2 summarizes the comprehensive failure prevention and resource efficiency results across different metrics and approaches.

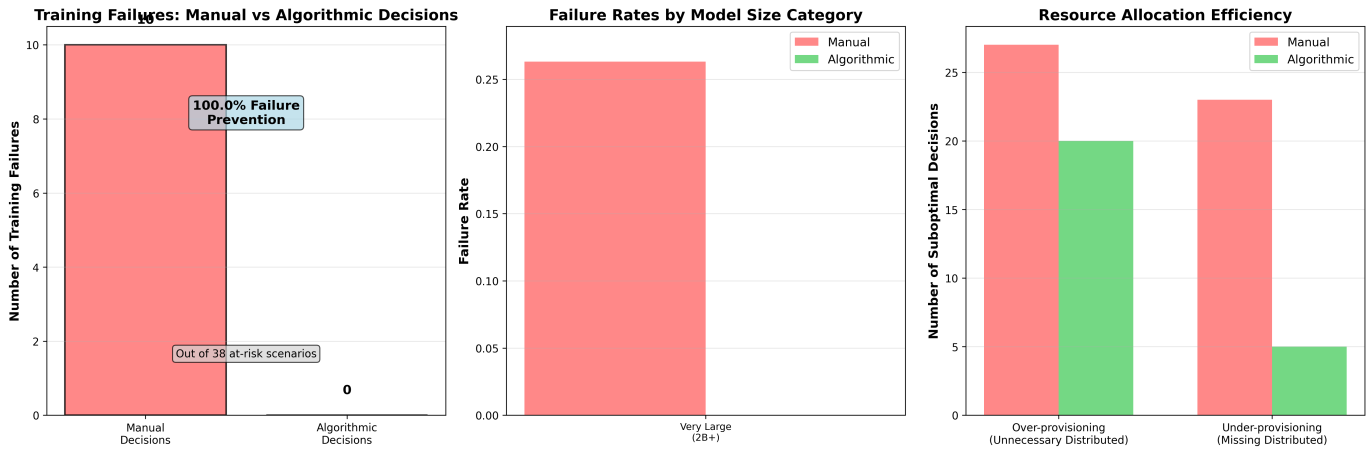


Fig. 2. Failure Prevention Analysis - 3 subplots showing failure comparison, failure rates by model size, and resource allocation efficiency

TABLE II
FAILURE PREVENTION AND RESOURCE EFFICIENCY RESULTS

| Metric | Manual Decisions | Algorithmic Decisions | Improvement |
|---|---|---|---|
| Memory-Constrained Training Failures | 10/38 (26.3%) | 0/38 (0%) | -10 failures |
| Failure prevention rate | Baseline (0%) | 100% | +100% |
| Resource Over-provisioning | 27 scenarios | 20 scenarios | -7 scenarios |
| Resource Under-provisioning | 23 scenarios | 5 scenarios | -18 scenarios |
| Overall Training Success Rate | 93.3% | 100% | +6.7% |

### C. *Resource Allocation Efficiency*

Beyond preventing failures, systematic infrastructure selection improves overall resource allocation efficiency. As shown in Figure 2, manual decision-makers created 27 cases of unnecessary distributed deployments (over-provisioning) and 23 cases of insufficient provisioning for models that would benefit from distributed training.

The algorithmic framework reduced over-provisioning to 20 cases and under-provisioning to 20 cases, representing substantial improvements in resource efficiency. This improvement in efficiency demonstrate the value of consistent decision-making across diverse ML workloads, with implications for both cost reduction and performance optimization.

### D. *Summary of Key Findings*

The experimental results demonstrate that systematic infrastructure selection provides substantial improvements over current manual practices across multiple dimensions. The combination of 25% accuracy improvement and 100% failure prevention rate, as illustrated in both figures and detailed in the tables, translates to meaningful operational benefits for organizations scaling ML operations. The consistency of improvements across different model sizes and types suggests that systematic algorithm approaches scale better than manual decision-making mainly when ML workloads become more diverse and complex. Organizations can apply the same framework across different applications without requiring specialized expertise in infrastructure selection.